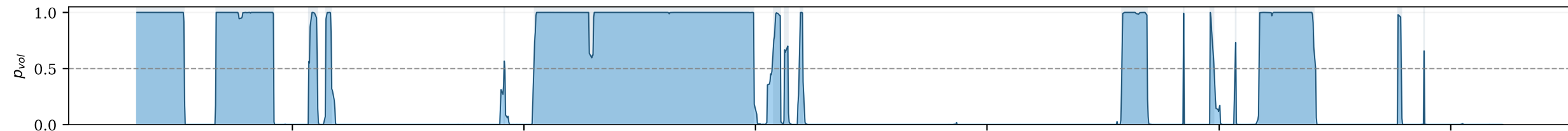
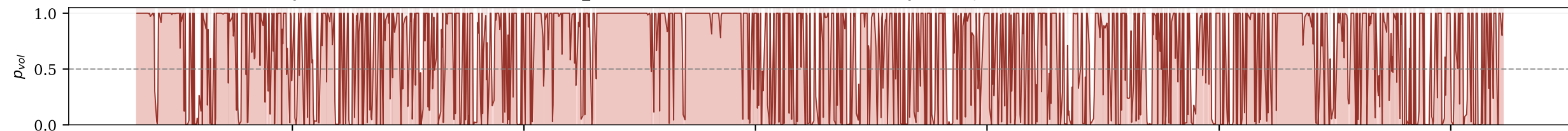


HMM-posterior p_{volatile} over the test period — shaded bands mark contiguous days with $p_{\text{vol}} > 0.5$

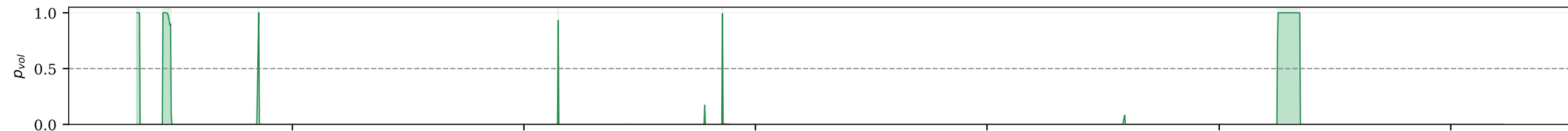
Variant O HMM — refit as plain GaussianHMM (n_mix=1, price/vol only, 11 features) — 33% of test days have $p_{\text{vol}} > 0.5$



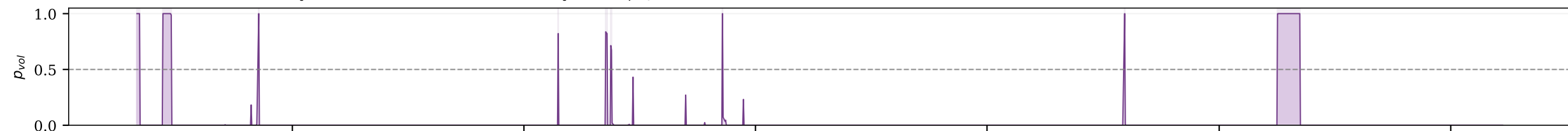
Variant O HMM — currently saved (BIC winner: GMMHMM, n_mix=2, 11 features) — 67% of test days have $p_{\text{vol}} > 0.5$



Variant A HMM (+AAII sentiment, 13 features) — 3% of test days have $p_{\text{vol}} > 0.5$



Variant B HMM (+VIX-family, 16 features) — 3% of test days have $p_{\text{vol}} > 0.5$



2021

2022

2023

2024

2025

2026

Date (test period)